

Lab Resource Utilization Platform

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Project Understanding and Objectives

Infosys Springboard Internship

1. Introduction

The Lab Resource Utilization Platform is a web-based application designed to help educational institutions manage laboratory resources efficiently. The system enables administrators, laboratory managers, faculty members, researchers, and students to monitor equipment availability, reserve laboratory resources, schedule maintenance, and generate utilization reports. The platform minimizes manual work and improves transparency in laboratory operations.

2. Project Overview

Many laboratories rely on spreadsheets and paper registers to manage equipment and bookings. This often results in scheduling conflicts, underutilized resources, missing maintenance records, and inefficient reporting. The proposed platform provides centralized management of laboratory assets through a secure, role-based web application.

3. Project Objectives

- Develop a centralized laboratory resource management system.
- Enable online equipment booking and approval.
- Track equipment utilization in real time.
- Maintain maintenance schedules and service history.
- Support inter-department and inter-institution resource sharing.
- Generate utilization and maintenance reports.
- Improve transparency, efficiency, and resource utilization.

4. Project Scope

The system covers user authentication, laboratory equipment management, booking management, maintenance tracking, utilization monitoring, notifications, reporting, and role-based access control. It is intended for colleges, universities, research centers, and similar institutions.

5. Stakeholders

Stakeholder	Responsibility
System Administrator	Manage the entire platform.
Institution Administrator	Manage institution-level resources.
Lab Manager	Manage equipment and bookings.
Lab Technician	Maintain equipment records.
Faculty/Researcher	Reserve and use equipment.
Student	Request and use available resources.

6. Expected Outcomes

- Efficient utilization of laboratory equipment.
- Reduced booking conflicts.
- Improved maintenance planning.
- Accurate reporting and analytics.
- Better user experience through automation.

7. Key Features

- Secure Login and Role-Based Access Control
- Equipment Management
- Online Equipment Booking
- Maintenance Scheduling
- Resource Sharing
- Notifications
- Reports and Analytics

8. Benefits

The platform reduces manual effort, improves laboratory efficiency, increases equipment availability, enhances transparency, and provides reliable data for decision-making.

9. Conclusion

The Lab Resource Utilization Platform offers a scalable and user-friendly solution for managing laboratory resources. By automating equipment booking, maintenance, and reporting, the platform helps institutions maximize resource utilization while providing a better experience for administrators and users.